MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 11746252 POWER CONTROL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirement for one type of circuit card assembly. The requirements of this specification together with the general requirements of Specification MIL-C-48411 comprise the complete requirements for Power Control.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-48411	Circuit Card Assemblies, General Specification for

DRAWINGS

11746252	Circuit Card Assembly, Power Control

(Copies of specifications, standards, drawings, and publications required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item Identification.- This assembly is a single ended printed circuit board consisting of resistive divider networks, operational amplifiers, voltage regulator, and a synchro converter to provide power control of the simulator and synchro demodulation information.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 11746252.

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3.2 Characteristics.

3.2.1 Performance.- The performance requirements specified herein shall be met on every assembly at standard ambient conditions prior to Government acceptance.

3.2.1.1 Power supply.- The requirements of 3.2 shall be met with the following power supply voltages.

a. +10 VDC
b. +15 VDC
c. -15 VDC
d. +28 VDC
e. 115 VAC 400 Hz

3.2.1.2 Low Voltage Indicator.- The low voltage indicator circuit shall be capable of indicating the presence of 24 VDC.

3.2.1.3 Test Voltages.- The test voltages for pins in Column B shall be as shown in Column C of Table I.

TABLE I

COLUMN A	COLUMN B	COLUMN C
PIN	PIN	VOLTAGE (DC)
22	11	3.741 ± 50 mV
16	13	5.000 ± 50 mV
19	14	6.671 ± 50 mV
20	15	5.656 ± 50 mV
29	24	5.000 ± 50 mV
30	26	-3.741 ± 50 mV
25	31	19.971 ± 200 mV
37	32	5.000 ± 50 mV
28	33	-3.333 ± 50 mV

3.2.1.4 +10 and -10 Volt Output.- The output voltage on pins 16 and 28 shall be plus and minus 10 VDC respectively.

3.2.1.4 Synchro Output.- The synchro shall be capable of incremental voltage changes of from plus 10 to minus 10 volts DC.

3.2.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-G-48411.

3.2.3 Physical Characteristics.- The physical characteristics shall be as specified on drawing 11746252 and the engineering parts list.

3.3 Design and Construction.

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3.3.1 Production Drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-G-48411, Drawing 11746252 and associated lists.

3.3.2 Standard of Manufacture.- The standard of manufacture shall be those specified in specification MIL-G-48411.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-G-48411.

3.3.4 Preproduction sample.- Preproduction sampling shall be as specified in specification MIL-G-48411.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-G-48411.

4.1.2 Special test and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-G-48411.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-G-48411.

4.2.1 Test equipment.- (see Figure 1)

4.2.1.1 Power Supply (PS^1).- The power supply shall be 15 ± 0.001 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.2 Power Supply (PS 2 & PS 3).- The power supply shall be 20 ± 1.0 VDC, 100 mA or greater and a ripple of less than 10 mV RMS.

4.2.1.3 Power Supply (PS 4).- The power supply shall be 40 ± 0.5 VDC, 200 mA or greater and a ripple voltage of less than 20 mV RMS.

4.2.1.4 Power Supply (PS 5).- The power supply shall be 115 ± 5 VAC 400 ± 20 Hz.

4.2.1.5 Dummy Loads.- The dummy loads shall be a MS 25237-327 miniature bulb or its equivalent and resistance of 10.000 ± 0.01% ohm, 1/2 watt as shown in Figure 1.

4.2.1.6 Digital Voltmeter.- The digital voltmeter shall have a voltage accuracy and resolution of 0.1 percent or better from zero to 100 VDC. Input inpedance of 10 megohms or greater on all voltage scales.

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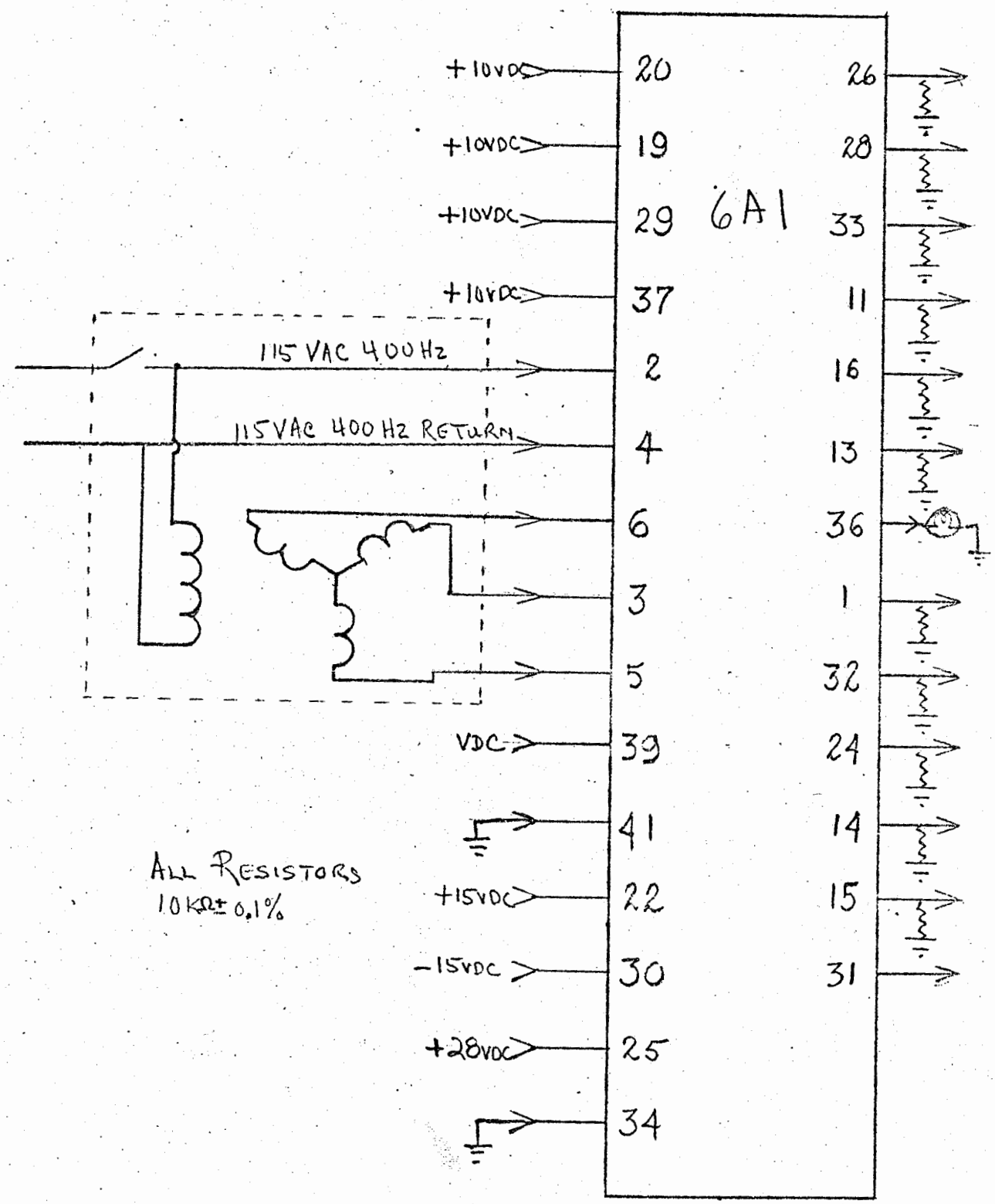


FIGURE I

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4.2.2 Performance.- Unless otherwise specified, the performance tests specified herein shall be satisfactorily met at standard ambient conditions. Standard ambient conditions for the purpose of these tests shall be defined as:

a. Temperature between 60°F and 90°F.
b. Barometric pressure between 28 and 32 inches of mercury.
c. Relative Humidity (RH) up to 90%.

4.2.2.1 Power Supply.- All the requirements shall be met with the following power supply voltages.

a. +10.000 VDC ± 1 mv
b. +15.000 VDC ± 1 mv
c. -15.000 VDC ± 1 mv
d. +28.000 VDC ± 1 mv
e. 115 ± 5.0 VAC, 400 ± 20 Hz

4.2.2.2 Test Set-up.- The test of the circuit card assembly shall be performed with the inputs and outputs as diagrammed in Figure 1 unless otherwise specified.

4.2.2.3 Low Voltage Indicator.- When 30 VDC is applied to pin 39, the lamp connected to pin 36 shall light. When the voltage on pin 39 is reduced to 24 ± 0.2 V the lamp shall blink.

4.2.2.4 Test Voltages.- When voltages as shown in Figure I are applied to the pins in column A of Table I, the voltage on the corresponding pins in column B with respect to ground shall be as shown in column C.

4.2.2.5 +10 Volt Power Test.- With +15 VDC applied to pin 22, R14 shall be capable of adjusting the output at pin 16 so as to obtain 10.000 ± 0.001 VDC.

4.2.2.6 -10 Volt Power Test.- With -15 VDC applied to pin 30, R17 shall be capable of adjusting the output of pin 28 so as to obtain 10.000 ± 0.001 VDC.

4.2.2.7 Synchro Output.- With the test synchro connected as shown in Figure I, a plus and minus 35° rotation of the synchro control shall cause the output at pin 1 to tranverse smoothly between the limits of plus and minus 10 VDC.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the requirements of Packaging Data Sheet (PDS), AMC Form 1029, 11746252.

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6. NOTES

6.1 The notes specified in the general specification MIL-G-48411 are applied to this specification.

Custodian:

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Preparing Activity:

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Project No. 1240-A596

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